

Pierre Chamoun

 [pierre-chamoun](#) |  [pierre-chamoun](#) |  [pierre-chamoun.github.io](#) |  pierre.chamoun@mail.mcgill.ca

EDUCATION

McGill University

Expected Dec. 2027

Master of Science (M.Sc.), Robotics

GPA: 4.0/4.0 | FRQNT Master's Scholarship | MEUSMA

McGill University

Dec. 2025

Bachelor of Engineering (B.Eng.), Mechanical Engineering

GPA: 3.97/4.00 | J.W. McConnell Scholarship | NSERC Research Award | Dean's Honour List | McGill Robotics

TECHNICAL SKILLS

Programming	C++, Python, MATLAB
Robotics & Tools	ROS, GTSAM, Docker, Git
Technical Areas	State Estimation, Control Theory, Computer Vision, Machine Learning
Other	Scientific computing, Statistics, GD&T, CAD

WORK EXPERIENCE

DECAR

Jan 2026 – Present

Graduate Researcher in Robotics

- Investigating targetless multi-sensor calibration in unstructured environments for autonomous mobile robots.

Pratt & Whitney Canada

Apr 2025 – May 2025

Internship in Thermal Analysis and CFD, Turbine Rotating Structures

- Recreated a CAD model in CATIA of a rotating heat transfer test rig originally used in the 1980s to develop one-dimensional correlations for heat transfer coefficients on rotating turbine discs.
- Developed a CFX computational fluid dynamics (CFD) model of rotating turbine discs and calibrated it to match heat-transfer measurements obtained from thermocouples at specific locations.
- Derived three-dimensional heat-transfer correlations from the CFD simulation results.

Pratt & Whitney Canada

Jan 2025 – Apr 2025

Engineering Internship in Mechanical Design

- Introduced engineering changes to reduce non-conformities resulting from the manufacturing process or to accommodate supplier requests.
- Validated and prepared documentation for various engineering change requests.
- Familiarized with the operation, manufacturing, and assembly of Pratt & Whitney Canada's main gas turbine engines.

McGill Aerodynamics Laboratory

May 2024 – Aug 2024

Research Internship in Computational Fluid Dynamics

- Applied machine learning techniques to study a hydrodynamic field resulting from a simulation of a turbulent oscillatory internal flow.
- Used Python to perform principal component analysis (PCA) to reduce the dimensionality of the information space needed to describe the hydrodynamic field.
- Calibrated a nonlinear model to predict the temporal dynamics of the fluid velocity field by estimating the evolution of its principal components.

McGill Aerodynamics Laboratory

May 2023 – Aug 2023

Research Internship in Computational Fluid Dynamics

- Conducted direct numerical simulations using the C++ open-source software Channelflow to study turbulent internal flows subjected to an oscillatory pressure gradient.
- Used the Unix interface to operate the Graham computing cluster of the Canadian Digital Research Alliance.
- Performed data analysis with Python using statistical and spectral methods.